AN H SHEAF IS INSENSITIVE TO FINITE PRESENTATION THICKENINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVE (sources/stacks-flat/pinned-original-source.tex line 11259).

Allowed standard prerequisites (editorial).

- Surjective proper finite presentation morphisms are h coverings and the sheaf property. Same chapter, Lemma labelled lemma-surjective-proper-finite-presentation-h, cited at proof lines 11270-11274. Supplies the covering for the first proof.
- The cartesian property for almost blowup squares. Same chapter, Lemma labelled lemma-blow-up-square-h, cited at proof lines 11286-11288. Supplies the conclusion in the alternative proof.
- Vanishing of the affine blowup algebra in a nilpotent element. Algebra, Section labelled algebra-section-blow-up, cited at proof lines 11276-11280. Shows that the blowup of X' in X is empty, in the alternative proof.

Source reference. Original definition 0EU1: sources/stacks-flat/pinned-original-source.tex, source lines 10504–10504. Same-chapter pointer into the author environment definition at source line 10504; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ETW: sources/stacks-flat/pinned-original-source.tex, source lines 10374–10374. Same-chapter pointer into the author environment lemma at source line 10374; invoked at proof line 11271 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 052P: sources/stacks-chow/tags.txt, source lines 6577–6577. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11277 of the counted proof.

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Source reference. Original lemma 0EVD: sources/stacks-flat/pinned-original-source.tex, source lines 11216–11216. Same-chapter pointer into the author environment lemma at source line 11216; invoked at proof line 11288 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $\mathcal{F}$ be a sheaf on one of the sites $(Sch/S)_h$ constructed in Definition 0EU1. Let $X \to X'$ be a morphism of $(Sch/S)_h$ which is a thickening and of finite presentation. Then $\mathcal{F}(X') \to \mathcal{F}(X)$ is bijective.*

Proof. First proof. Observe that $X \to X'$ is a proper surjective morphism of finite presentation and $X \times_{X'} X = X$. By the sheaf property for the h covering $\{X \to X'\}$ (Lemma 0ETW) we conclude.

Second proof (silly). The blow up of X' in X is the empty scheme. The reason is that the affine blowup algebra $A[\frac{I}{a}]$ (Algebra, Section 052P) is zero if a is a nilpotent element of A. Details omitted. Hence we get an almost blow up square of the form

$$\begin{array}{ccc} \emptyset & \longrightarrow & \emptyset \\ \downarrow & & \downarrow \\ X & \longrightarrow & X' \end{array}$$

Since $\mathcal{F}$ is a sheaf we have that $\mathcal{F}(\emptyset)$ is a singleton. Applying Lemma 0EVD we get the conclusion. □